
The Verification Gap: What Happens When Checking Costs More Than Making

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Abstract

Most arguments about AI risk turn on whether machines will deceive us, but I think the nearer disaster is a duller one that is already underway: AI is exhausting our capacity to verify anything at all. This essay takes the four things researchers currently mean by “epistemic collapse” (model collapse, validation overload, the mistaking of statistics for knowledge, and societal fragmentation) and argues that they are really one cascading failure with a single root cause, namely that producing a claim has become almost free while checking one still costs what it always did. Alignment does not touch this problem even if we solve it perfectly, and the most serious governance proposals in circulation devote hundreds of pages to compute verification while giving epistemics a couple of paragraphs. Verification capacity is the binding constraint on everything else, and almost nobody is building for it.

1 The wrong question

Ask someone worried about AI what keeps them up at night and you will usually get a story about deception, whether that means a model that schemes or a superintelligence that tells its evaluators exactly what they want to hear until the day it no longer needs to, and since serious people spend their careers on those questions, nothing in this essay is meant to wave them away.

What I want to point at instead is a quieter failure mode that needs no scheming, no superintelligence, and no malice, because all it needs is volume. In January 2026, researchers showed people AI-generated deepfake videos of crimes while telling them, explicitly and in advance, that the videos were fake, and the warning barely mattered, since even participants who consciously accepted it remained influenced by what they had seen (Sage, 2025). What that experiment reveals is not that deepfakes are convincing but something stranger and worse, which is that knowing a thing is fake no longer reliably protects you from it, because the machinery in your head that turns perception into belief evolved for a world where seeing something meant it had probably happened, and was never built for an environment where most of what reaches your eyes might be manufactured.

When you scale that single finding up to the size of the internet, the picture gets grim quickly. By mid-2025 more than 1,200 AI-generated “news” sites were publishing under plausible mastheads in sixteen languages, the EU was counting AI in 27% of foreign information-manipulation attempts (nearly triple the previous year’s share), and AI-generated misleading posts were going disproportionately viral despite mostly originating from small accounts (arXiv 2505.10266,

2025), all without a single misaligned model anywhere in the pipeline, because the only thing any of it required was that generating content became cheap.

The danger, as one line of this research puts it, is not that people will be fooled but that they will stop trying to figure out what is true, and not because they are gullible but because they are exhausted.

2 Four things “epistemic collapse” means

The term “epistemic collapse” is now in use across at least four research communities that do not quite mean the same thing by it, and it is worth pulling the meanings apart carefully, because the real claim of this essay is that they describe stages of a single cascade rather than four separate problems.

1. Technical: model collapse. Train a model on the outputs of a previous model, repeat the process a few times, and the tails of the original data distribution vanish while each generation drifts further from the ground truth it was once anchored to. This is not conjecture, since it has been proven statistically that collapse cannot be avoided when training solely on synthetic data (arXiv 2404.05090, 2024) and shown empirically that the effect extends beyond text into vision-language and diffusion models (arXiv 2505.08803, 2025), though there is a partial reprieve in that mixtures which keep the synthetic fraction below some maximal threshold can remain stable even while the pure recursive case stays mathematically doomed.

The ecosystem version of this story turns out to be worse than the single-model version. When researchers tested 27 LLMs across 155 topics they found every one of them less epistemically diverse than a basic web search, with larger models producing less diverse claims rather than more (arXiv 2510.04226, 2025), and follow-up work on simulated ecosystems found that homogeneity actively accelerates the rot, since “scaling up the system can amplify collapse in highly homogeneous ecosystems” (arXiv 2512.15011, 2025), which means monoculture is not just a market concern but an epistemic accelerant.

2. Infrastructural: validation overload. Singh reframes the whole problem as a structural failure of validation architecture rather than a misinformation problem (Singh, 2026), driven by three mechanisms that feed each other: epistemic inflation, in which far more claims are produced than anyone has the capacity to verify; recursive drift, in which synthetic content feeds on itself and slides away from empirical reality; and validation fatigue, in which the humans and institutions doing the checking simply wear down under impossible load. The observation underneath all three is that peer reviewers, fact-checkers, editors, courts, and teachers are rate-limited systems trying to keep pace with a generation process that is no longer rate-limited by anything human at all.

3. Epistemological: misrecognition. Alex Reid borrows Virilio’s principle that every technology invents its own accident and argues that generative AI’s accident is epistemic collapse itself, the state in which “there is no shared process of verification in a community” (Reid, 2026). His sharpest point concerns what model outputs actually are: since a model neither knows nor claims to know the content of what it produces, its outputs are necessarily non-epistemic, and “epistemic content collapses into statistical distribution.” When a reader treats such output as

a knowledge claim anyway, the reader is performing the epistemics on the model’s behalf, quietly lending their own credibility to something that has none of its own, and once you multiply that small act of misrecognition by billions of daily interactions you arrive at a society that systematically miscounts how much verified knowledge it actually possesses.

4. Social: fragmentation. Downstream of the other three sits the version that makes the news, the world of disconnected information realities, lost common ground, and radicalization inside silos. This layer usually gets treated as its own phenomenon with its own literature, something like a social-media problem or a polarization problem, but I think it is better understood as the surface expression of everything underneath it.

Seen that way, the cascade runs in order: technical collapse degrades the tools, which overwhelms and erodes the validation infrastructure, which normalizes misrecognition because unverified content is increasingly all there is, which finally fragments the shared reality that deliberation depends on. What look like four literatures are describing one failure.

3 Industrialization is the difference

The obvious pushback here is that misinformation is ancient, that forged letters, yellow journalism, and wartime propaganda are as old as their respective media, and that humanity has never lived in an epistemically clean environment anyway, which is why we evolved journals, courts, editorial desks, and peer review to cope.

But what is new is not deception, it is the unit economics of deception. Every coping institution we have quietly assumes that producing a credible-looking claim costs something, whether that cost is paid in time, expertise, reputation, or printing presses, and verification could therefore afford to be slower than generation because generation was expensive too. That assumption has now simply stopped being true, since producing a plausible claim costs approximately nothing while checking one costs what it always did, and recent work names the result “industrialized deception,” the automated production of misleading content at a scale no verification system was ever engineered for (arXiv 2601.21963, 2026).

Ferrara’s “generative AI paradox” describes where this ends up: as synthetic media becomes indistinguishable from authentic media, the individually rational move is to discount all digital evidence rather than sort the true from the false, which produces not a world full of successful lies but a world where evidence stops working altogether. In economic terms this is a market failure, and the right analogy is not counterfeiting but debasement, because what loses value is the currency itself.

There is also a feedback loop here, and it has been formalized. If you model humans and language models as one coupled dynamical system linked by usage, generation, and retraining, you find three regimes, running from co-evolutionary enhancement through fragile equilibrium down to degenerative convergence (arXiv 2605.06347, 2026), and the failure mode is a slide into a low-diversity equilibrium in which model quality and human cognition degrade together. The authors give this a name worth remembering, “dynamical misalignment,” because no individual component is misaligned and the system fails anyway, and that property matters enormously for policy, since you cannot fix a systemic failure by certifying the components.

4 Why the alignment frame misses it

It would be unfair to say the AI-risk community ignores epistemics entirely, because the most detailed governance proposal currently on the table, the AI Futures Project’s *AI 2040* scenario with its “Plan A” for an internationally verified slowdown, is admirably concrete about a remarkable number of things (AI Futures Project, 2026), specifying optical network taps for inference-only datacenter verification, designing compute declarations audited across the entire chip supply chain, and inventing mutually assured compute destruction as a deterrence backstop, with hundreds of pages of supplements modeling takeoff speeds, covert-project detection odds, and permit economics.

On epistemics, though, the same document offers a two-page appendix, and while that appendix names a genuinely useful idea in the “basin of sanity,” a self-reinforcing equilibrium in which a society sane enough to adopt truth-seeking AI tools becomes saner over time, set against a vicious loop in which sycophantic AI degrades collective judgment, it then gestures at a handful of interventions (honest-AI evaluations, privacy-preserving auditing, lie detectors, better forecasting) and moves on. The basin is asserted rather than engineered, and nothing in those two pages tells us how a society enters it, how we would measure our distance from its boundary, or what keeps it stable against actors who profit from the other basin.

None of this is a knock on the authors, who at least flagged the problem when most proposals never mention it, but it does reveal the field’s priority ordering, whose working assumption seems to be that if we control the compute and align the models, the epistemic environment will mostly take care of itself.

The research above says otherwise, because a perfectly aligned model still floods the commons. Alignment constrains intent, and the cascade does not run on intent, it runs on volume, which means honest models trained on an increasingly synthetic corpus still drift, validators still burn out under content produced in complete good faith, and readers still mistake statistical output for knowledge even when the statistics are benign. Alignment is necessary for the deception problem while being nearly irrelevant to the exhaustion problem.

Underneath that priority ordering sits a deeper structural mismatch. Alignment is a property of an artifact, which means somebody can own it, whether that is a lab, a safety team, or a regulator certifying a model before release, whereas verification capacity is a property of an ecosystem, living in the coupling between models, institutions, incentives, and habits of mind. Our institutions are good at assigning responsibility for artifacts and notably bad at assigning responsibility for ecosystems, so the tractable, ownable problem absorbs the funding and the talent while the binding constraint absorbs neither.

5 What building for verification would look like

If verification capacity really is the constraint, the design goal follows directly: engineer the epistemic infrastructure so that verification scales with generation, rather than hoping that institutions built for human-speed publishing somehow survive machine-speed publishing. The pieces of a research program already exist scattered through the literature, and what is missing is anyone treating them as one agenda.

1. **Instrument the commons.** Singh’s framework ships with diagnostic signals, including resilience gradients, drift divergence exceeding adjudication resolution, and validator-fatigue

thresholds (Singh, 2026), while the dynamical-systems work identifies the transition from fragile equilibrium to degenerative convergence as the event to watch for (arXiv 2605.06347, 2026). Since nobody currently computes these numbers for any real knowledge domain, an early-warning observatory for epistemic health, one that tracked claim-to-verification ratios, diversity indices, and validator backlogs across science, journalism, and law, would be both feasible with today’s tools and entirely novel.

2. **Mandate diversity where monoculture accelerates collapse.** If homogeneous model ecosystems demonstrably amplify degradation (arXiv 2512.15011, 2025), then model diversity stops being an antitrust nicety and becomes an epistemic safety property with an optimal level that can in principle be estimated per domain, which would reframe a meaningful chunk of AI governance in measurable terms.
3. **Distribute epistemic authority.** Proposals like open cognitive graphs, which externalize the conceptual structure of AI knowledge so that it can be inspected, contested, and revised, point at the right target of keeping the authority to say “this is known” distributed and revisable rather than consolidated inside a handful of opaque systems (arXiv 2602.16949, 2026), and provenance belongs at the same layer, not as watermarking bolted on after the fact but as chains of epistemic custody treated as a first-class property of published content.
4. **Fund verification the way we fund generation.** The asymmetry is ultimately economic, so it can be attacked economically, and while nearly all AI investment currently subsidizes the generation side of the ledger, there is no structural reason capital could not flow toward the checking side instead, into automated fact-verification, reproducibility tooling, and institutional adjudication capacity, except that nobody has framed that side as the complement the market is missing.

Honesty requires saying that these are directions rather than solutions, since the optimal-diversity question is open, whether early-warning signals generalize across domains is open, and how epistemic authority should be distributed without the distribution itself being captured is very open. But that openness is exactly the point, because it means these problems are tractable, measurable, and largely unclaimed.

6 Nobody owns this problem

Here is the strange sociological fact that motivated this essay: the machine learning community has proven collapse theorems, the AI-ethics community has formalized validation failure, philosophers of technology have diagnosed the misrecognition, social scientists have measured the fragmentation, and the most ambitious AI governance proposal in circulation concedes, in an appendix, that keeping society inside the “basin of sanity” should be a top governmental priority during AI takeoff, and yet there is no field here. There are no shared benchmarks, no observatory, no funding stream, and no institution anywhere whose mandate is the verification capacity of the knowledge commons, so while alignment has labs, fellowships, and conferences, epistemic infrastructure has scattered arXiv postings that only occasionally cite across disciplinary lines. The problem has been named convergently by four different communities, and although naming is usually the step right before a field crystallizes, this one has not crystallized.

That is the opportunity, and it is also the warning, because with every year the generation-verification gap widens, the institutions that would need to build this infrastructure operate with less credibility and more fatigue than the year before. Epistemic collapse is the rare risk that erodes our ability to respond to it as a direct consequence of the risk materializing, since a society that can no longer agree on how to verify claims cannot verify the claim that it should fix its verification systems.

The race to build more capable AI has a thousand entrants, while the race to keep truth checkable has, so far, almost none, and it is worth asking which of the two is actually the bottleneck on everything else.

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